Annex C. RabbitMQ events and queues

Version 1.0 · October 28, 2026. The services notify each other through events on RabbitMQ 4.1 with a *transactional outbox*: the message row is written in the same transaction as the change, a relay publishes it every second with broker confirmation, delivery is at least once, the queues are durable and each has its `.dlq` (*dead-letter* after three attempts). An *outbox* stops at a message that no queue receives, which is why the consuming service is deployed before the producer.

Events, producers and queues

Event	Producer	Consuming queue	Consumer
`measurement.recorded`	measurements	`alert-service.measurement.recorded`	alerts
`device.connectivity-lost`, `device.connectivity-restored`	devices	`alert-service.device.events`	alerts
`device.linked`, `device.unlinked`	devices	`alert-service.device.events` and `notification-service.notification.activity`	alerts and notifications; measurements use them for their cache
`alert.raised`	alerts	`notification-service.notification.alert-raised`	notifications (also the unusual consumption alert, key `alert.anomaly.device`)
`recommendation.created`	recommendations	`notification-service.notification.recommendation-created`	notifications
`account.registered`, `account.administered`	security	`notification-service.notification.account-events`	notifications
`account.deleted`	security	`home-service.home.account-deleted` and `notification-service.notification.account-events`	homes and notifications (cleanup)
`mail.email-verification`, `mail.password-reset`, `mail.account-deletion`	security	`notification-service.notification.mail`	notifications (the only one that sends email)
`home.created`, `home.member-joined`, `home.member-left`, `home.member-removed`, `home.thresholds-updated`, `home.ownership-transferred`	homes	`notification-service.notification.activity`	notifications
`home.deleted`	homes	`notification-service.notification.activity` and the device service's queue	notifications and devices (unlinks the modules and sends `unlink`)

Switches of the consumers and relays

All are `true` by default and exist only to roll back during a migration: `HOME_OUTBOX_RELAY_ENABLED`, `ACCOUNT_DELETED_CONSUMER_ENABLED`, `SECURITY_EVENTS_CONSUMER_ENABLED`, `ALERT_EVENTS_CONSUMER_ENABLED`, `ACTIVITY_CONSUMER_ENABLED`, `ALERT_CONSUMERS_ENABLED`, `ALERT_SERVICE_OUTBOX_RELAY_ENABLED`, `RECOMMENDATION_SCHEDULER_ENABLED` and `RECOMMENDATION_OUTBOX_RELAY_ENABLED` (Annex B).

Retired queues

With the backend retirement on October 28, 2026 the ten `energy-monitor-backend.*` queues (each with its `.dlq`) were deleted with `rabbitmqctl delete_queue`: they no longer had a consumer.

Check

```
emc exec rabbitmq rabbitmqctl list_queues name messages     # the .dlq queues must stay at 0
```

Spanish version: ../ES/anexo-c-eventos-y-colas.md